Mathematical Foundations of Machine Learning

Week 4: Spectral Theory

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1. Eigenvalues and eigenvectors
2. Diagonalization
3. Spectral theorem
4. Geometric interpretation
5. Schur decomposition

Quick Recap — Week 3

Definition 0.1. A map $T : V \to W$ between vector spaces is a **linear transformation** if

$$T(u+v) = T(u) + T(v), \qquad T(\alpha v) = \alpha T(v)$$

for all $u, v \in V$ and $\alpha \in \mathbb{R}$.

Example.
- ▶ $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (2x, 3y)$.
- ▶ Derivative $D : C^1([0,1]) \to C([0,1])$, $Df = f'$.

Matrix Representation

Every $T : \mathbb{R}^n \to \mathbb{R}^m$ is determined by its action on the basis vectors:

$$T(e_j) = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}, \qquad T(x) = Ax, \quad A = (a_{ij}) \in \mathbb{R}^{m\times n}.$$

So linear transformations $\leftrightarrow$ matrices (relative to chosen bases).

Example. $T(x, y) = (x + y, x - y)$. Then

$$T(e_1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Geometric View

Linear transformations capture:

- Scaling: $T(x, y) = (2x, 2y)$.
- Rotation: $T(x, y) = (\cos\theta\, x - \sin\theta\, y,\ \sin\theta\, x + \cos\theta\, y)$.
- Projection: $T(x, y) = (x, 0)$.

Change of Basis

Given another basis $B = \{v_1, \dots, v_n\}$ with change-of-basis matrix P, if A is the standard matrix of T, then the matrix in basis B is

$$A_B = P^{-1}AP.$$

Example. For $T(x, y, z) = (x + 2y, y + z, x - z)$, in the standard basis

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}.$$

With $B = \{(1, 0, 0), (1, 1, 0), (0, 0, 1)\}$, the basis matrix is

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A_B = P^{-1}AP.$$

1 Representation Theorem

Definition 1.1. A **linear functional** on a vector space V is a linear map

$$f : V \to \mathbb{R}.$$

That is, for $u, v \in V$ and $\alpha \in \mathbb{R}$,

$$f(u+v) = f(u) + f(v), \qquad f(\alpha v) = \alpha f(v).$$

Example.
- On $\mathbb{R}^n$, $f(x_1, \dots, x_n) = 2x_1 - 3x_2$ is linear.
- For polynomials $p(x)$, evaluation at $x = 1$ defines a linear functional $f(p) = p(1)$.

Recall that the set of all linear functionals on V is called the **dual space**, denoted V^*:

$$V^* = \{f : V \to \mathbb{R} \mid f \text{ linear}\}.$$

If V has dimension n, then $\dim V^* = n$ as well. Indeed, if $B = \{v_1, \dots, v_n\}$ is a basis of V, B determines a *dual basis* $B^* = \{f_1, \dots, f_n\} \subset V^*$ such that

$$f_i(v_j) = \delta_{ij}.$$

This provides coordinates for functionals just as B provides coordinates for vectors.

Example. For $V = \mathbb{R}^3$ with the standard basis $e_1 = (1,0,0)$, $e_2 = (0,1,0)$, $e_3 = (0,0,1)$, the dual basis consists of the coordinate projections:

$$f_1(x,y,z) = x, \quad f_2(x,y,z) = y, \quad f_3(x,y,z) = z.$$

If we choose a different basis, say $(1,1,0)^\top, (1,0,1)^\top, (0,1,1)^\top$, then the corresponding dual basis is

$$g_1(x,y,z) = x + y, \quad g_2(x,y,z) = x + z, \quad g_3(x,y,z) = y + z.$$

Given $T : V \to W$, there is an induced **dual map**

$$T^* : W^* \to V^*, \qquad (T^*g)(v) = g(T(v)).$$

Thus T^* acts on functionals by pre-composition. Further, if A is the matrix of T in bases B_V, B_W of V and W, then the matrix of T^* in the dual bases B_W^*, B_V^* is $A^\top$, the transpose.

Example. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x,y) = (x + 2y, y)$. Its standard matrix is

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad A^\top = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}.$$

So the dual map $T^* : \mathbb{R}^{2*} \to \mathbb{R}^{2*}$ has matrix $A^\top$.

Geometrically, vectors in V represent *directions*, while functionals in V^* represent *measurements* or *projections* onto $\mathbb{R}$. The dual space encodes how vectors are evaluated, and is fundamental in optimization, convexity, and regularization.

A central result in inner product spaces is that every linear functional can be "represented" by an inner product with a unique vector. This is known as the **Riesz Representation Theorem**.

Theorem 1.1 (Riesz Representation Theorem). *Let V be a finite-dimensional inner product space over $\mathbb{R}$. For every linear functional $f \in V^*$, there exists a unique vector $y \in V$ such that*

$$f(x) = \langle x, y \rangle \qquad \text{for all } x \in V.$$

Notice that :

- a linear functional $f : V \to \mathbb{R}$ is a linear "measurement" on vectors; as its domain the vector space V and its range is $\mathbb{R}$.
- The Riesz representation theorem asserts that such functionals are nothing more than inner products with some fixed vector $y \in V$ or conversely, to every measurement there corresponds a vector.
- The representing vector y is unique — which is very easy to see : if $\langle x, y_1 \rangle = \langle x, y_2 \rangle$ for all x, then $\langle x, y_1 - y_2 \rangle = 0$ for all x, which forces $y_1 = y_2$.

Proof. Choose an orthonormal basis $\{e_1, \ldots, e_n\}$ of V (we now that this exists by Gram Schmidt + finite dimensionality). For each j, define α_j as :

$$f(e_j) =: \alpha_j \in \mathbb{R}.$$

Now define

$$y = \sum_{j=1}^{n} \alpha_j e_j.$$

Then for any $x = \sum_j x_j e_j$,

$$\langle x, y \rangle = \Big\langle \sum_j x_j e_j,\ \sum_k \alpha_k e_k \Big\rangle = \sum_{j,k} x_j \alpha_k \langle e_j, e_k \rangle = \sum_j x_j \alpha_j.$$

On the other hand,

$$f(x) = f\left(\sum_j x_j e_j\right) = \sum_j x_j f(e_j) = \sum_j x_j \alpha_j.$$

Thus $f(x) = \langle x, y \rangle$ for all $x \in V$. ☐

Examples.

Example. Let $V = \mathbb{R}^3$ with the standard inner product, and $f(x, y, z) = 2x - 3y + z$. Then $f(e_1) = 2$, $f(e_2) = -3$, $f(e_3) = 1$, so the representing vector is $y = (2, -3, 1)$. Hence $f(v) = \langle v, (2, -3, 1) \rangle$.

Example. On $V = \mathcal{P}_3$ with inner product

$$\langle p, q \rangle = \int_0^1 p(t)q(t)\, dt,$$

define $f(p) = p(1)$ (evaluation at $t = 1$). By the Riesz Representation Theorem, for every continuous linear functional $L : \mathbb{P}_3 \to \mathbb{R}$, there exists a unique polynomial $r \in \mathbb{P}_3$ such that

$$L(p) = \langle p, r \rangle \qquad \text{for all } p \in \mathbb{P}_3.$$

We seek $r(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ such that

$$p(1) = \int_0^1 p(x)r(x)\, dx \qquad \text{for all } p \in \mathbb{P}_3.$$

This gives a system of equations obtained from the following :

$$\int_0^1 x^k r(x)\, dx = 1, \qquad k = 0, 1, 2, 3.$$

Writing $r(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$, we get

$$\frac{c_0}{k+1} + \frac{c_1}{k+2} + \frac{c_2}{k+3} + \frac{c_3}{k+4} = 1, \quad k = 0, 1, 2, 3.$$

Solving this linear system yields

$$c_0 = -4, \qquad c_1 = 60, \qquad c_2 = -180, \qquad c_3 = 140.$$

Thus, the representing polynomial is

$$\boxed{r(x) = -4 + 60x - 180x^2 + 140x^3.}$$

We may verifty the result : for any polynomial $p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \in \mathbb{P}_3$,

$$\begin{aligned}
\langle p, r \rangle &= \int_0^1 p(x)r(x)\, dx \\
&= a_0 \int_0^1 r(x)\, dx + a_1 \int_0^1 x r(x)\, dx + a_2 \int_0^1 x^2 r(x)\, dx + a_3 \int_0^1 x^3 r(x)\, dx \\
&= a_0 + a_1 + a_2 + a_3 = p(1).
\end{aligned}$$

Hence, the polynomial r represents the evaluation functional at $x = 1$ under the given inner product.

Geometric Note. The theorem tells us that the abstract space of linear functionals V^* can be identified with the vector space V itself, once an inner product is chosen. This identification is what underlies many duality principles in analysis and optimization.

1.1 Dual Norm

Norms measure the "size" of vectors, but they also give rise to a natural concept of duality. Once we have identified linear functionals with vectors via the inner product (Riesz Representation Theorem), it becomes meaningful to ask: what is the largest value a functional can attain on vectors of unit norm?

Definition 1.2 (Dual Norm)**.** Let $(V, \|\cdot\|)$ be a normed vector space. The **dual norm** on V^* is defined by

$$\|f\|_* = \sup_{\|x\|\le 1} |f(x)|, \qquad f \in V^*.$$

Notice that, if we identify f with a vector $y \in V$ via the Riesz Representation Theorem (so that V is finite dimensional) : $f(x) = \langle x, y\rangle$, then

$$\|y\|_* = \sup_{\|x\|\le 1} |\langle x, y\rangle|.$$

Key Properties.

- $\|\cdot\|_*$ is itself a norm on V^*.
- The Cauchy–Schwarz (or Hölder) inequality guarantees $|\langle x, y\rangle| \le \|x\|\,\|y\|_*$.
- Dual norms appear naturally in optimization and convex analysis: the value $\|y\|_*$ quantifies the maximal action of y against unit vectors.

Example. On $\mathbb{R}^n$ with the ℓ_1 norm, $\|x\|_1 = \sum_i |x_i|$, the dual norm is ℓ_∞: $\|y\|_\infty = \max_i |y_i|$. Indeed,

$$\sup_{\|x\|_1\le 1} |\langle x, y\rangle| = \sup_{\sum |x_i|\le 1} \left|\sum_i x_i y_i\right| = \max_i |y_i|.$$

Geometric Note. The unit ball of the dual norm is the polar set of the unit ball of the original norm:

$$B_* = \{y \in V : \langle x, y\rangle \le 1 \ \ \forall x \in B\},$$

where $B = \{x : \|x\| \le 1\}$. Thus, duality exchanges geometry of shapes (e.g. the ℓ_1 "diamond" and ℓ_∞ "square" are polars of one another).

Example (ℓ_1 vs ℓ_∞)**.** Let $y = (2, -5, 1)^\top \in \mathbb{R}^3$, and take the primal norm $\|\cdot\| = \|\cdot\|_1$. By definition,

$$\|y\|_* = \sup_{\|x\|_1\le 1} \langle y, x\rangle = \sup_{\sum_i |x_i|\le 1} \sum_i y_i x_i.$$

Upper bound: For any feasible x, $\sum_i y_i x_i \le \sum_i |y_i|\,|x_i| \le \|y\|_\infty \sum_i |x_i| \le \|y\|_\infty$. *Attainment:* Choose an index j with $|y_j| = \|y\|_\infty$ (here $j = 2$). Let $x^* = \operatorname{sgn}(y_j)e_j = -e_2$, which satisfies $\|x^*\|_1 = 1$ and $\langle y, x^*\rangle = |y_j| = \|y\|_\infty = 5$. Hence $\boxed{\|y\|_* = \|y\|_\infty = 5}$.

Example (ℓ_∞ vs ℓ_1)**.** With the same y but primal $\|\cdot\| = \|\cdot\|_\infty$,

$$\|y\|_* = \sup_{\|x\|_\infty\le 1} \langle y, x\rangle = \sup_{|x_i|\le 1} \sum_i y_i x_i.$$

Upper bound: $\sum_i y_i x_i \le \sum_i |y_i|\,|x_i| \le \sum_i |y_i| = \|y\|_1$. *Attainment:* Take $x^* = (\operatorname{sgn} y_1, \operatorname{sgn} y_2, \operatorname{sgn} y_3) = (1, -1, 1)$, which has $\|x^*\|_\infty = 1$ and $\langle y, x^*\rangle = |2| + |-5| + |1| = 8 = \|y\|_1$. Thus $\boxed{\|y\|_* = \|y\|_1 = 8}$.

Exercise.

- ▶ Show that on $\mathbb{R}^n$ with the ℓ_2 norm, the dual norm is again ℓ_2. *Hint:* $\|y\|_2 = \sqrt{\sum_i y_i^2}$. The one direction of the inequality follows directly from Cauchy–Schwarz:

  $$|\langle x, y\rangle| \le \|x\|_2\,\|y\|_2,$$

  and equality is attained for $x = \frac{y}{\|y\|_2}$.
- ▶ On $\mathbb{R}^n$ with the ℓ_∞ norm, $\|x\|_\infty = \max_i |x_i|$, the dual norm is ℓ_1: $\|y\|_1 = \sum_i |y_i|$.

2 Eigenvalues and Eigenvectors of Linear Transformations

So far, we have studied linear transformations $T : V \to V$ on a (finite-dimensional) vector space V. In general, such a transformation takes a vector and both changes its direction and scales its length. However, there are special vectors that are only scaled (possibly flipped) by T but not rotated into a new direction. These vectors are the *eigenvectors* of T. They play a very central role in understanding the geometry of such transformations.

Definition 2.1. Let $T : V \to V$ be a linear transformation on a vector space V. A nonzero vector $v \in V$ is called an **eigenvector** of T if there exists a scalar $\lambda \in \mathbb{R}$ such that

$$T(v) = \lambda v.$$

The scalar λ is called the **eigenvalue** corresponding to v.

Geometric meaning. Most vectors are altered in both direction and length by T. An eigenvector, however, is preserved in direction: applying T stretches (or shrinks) the vector by a factor of $|\lambda|$, and flips it if $\lambda < 0$. Thus eigenvectors describe *invariant directions* of the transformation.

Example. Consider $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(x, y) = (2x, 3y).$$

Then $(1, 0)$ and $(0, 1)$ are eigenvectors with eigenvalues 2 and 3, respectively, since

$$T(1, 0) = (2, 0) = 2(1, 0), \qquad T(0, 1) = (0, 3) = 3(0, 1).$$

Formally, v is an eigenvector with eigenvalue λ if and only if

$$T(v) - \lambda v = 0.$$

Equivalently,

$$(T - \lambda I)(v) = 0,$$

where I is the identity map on V. Hence, the eigenvalue problem is the search for scalars λ such that $T - \lambda I$ maps a non-zero vector v to the zero vector. This can only happen when $T - \lambda I$ has a nontrivial kernel.

Exercise. ▶ Say $T : V \to W$ is a linear map between two vector spaces V and W. Show that the following two sets are vector sub-spaces :

$$\ker(T) := \{v \in V \mid T(v) = 0\}$$
$$\operatorname{Im}(T) := \{T(v) \in W \mid v \in V\}.$$

▶ Show that a linear transformation $T : V \to W$ is one-to-one (injective) if and only if $\ker(T) = \{0\}$.

2.1 Determinants : A quick reminder

Geometrically, the determinant of a linear transformation measures how a it scales volume and whether it preserves or reverses orientation.

Definition 2.2. Let V be an n-dimensional real vector space and $\mathcal{B} = (v_1, \ldots, v_n)$ an ordered basis. For a linear map $T : V \to V$, write $[T]_\mathcal{B}$ for the matrix of T in the basis $\mathcal{B}$. The *determinant of* T is

$$\det(T) := \det\big([T]_\mathcal{B}\big).$$

Lemma 2.1 (Independence of Basis). *This definition is independent of the chosen ordered basis $\mathcal{B}$.*

Proof. Let $\mathcal{C}$ be another ordered basis and let P be the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{C}$. Then $[T]_\mathcal{C} = P^{-1}[T]_\mathcal{B}P$. Hence

$$\det\big([T]_\mathcal{C}\big) = \det(P^{-1})\,\det\big([T]_\mathcal{B}\big)\,\det(P) = \det\big([T]_\mathcal{B}\big).$$

□

When $V = \mathbb{R}^n$ with the standard inner product, $\det(T)$ is the oriented n-volume of the image of the unit cube under T. More generally, for any measurable $S \subset \mathbb{R}^n$,

$$\operatorname{Vol}_n\big(T(S)\big) = |\det(T)|\operatorname{Vol}_n(S).$$

Theorem 2.1 (Fundamental Properties). *Let $S, T : V \to V$ be linear and let $\lambda \in \mathbb{R}$. Then:*

(i) $\det(I_V) = 1$.

(ii) $\det(TS) = \det(T)\det(S)$.

(iii) *T is invertible* $\iff$ $\det(T) \neq 0$, *and in that case* $\det(T^{-1}) = \dfrac{1}{\det(T)}$.

(iv) $\det(\lambda T) = \lambda^n \det(T)$.

Choose any ordered basis $\mathcal{B}$ and compute $[T]_\mathcal{B}$. Then use your preferred matrix-determinant method (cofactor expansion, row reduction with sign/scale tracking, or LU) to obtain $\det(T) = \det([T]_\mathcal{B})$. By the lemma, the result does not depend on $\mathcal{B}$.

Example. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be given by $T(x, y) = (2x + y,\, 3x + 4y)$. With respect to the standard basis,

$$[T] = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}, \qquad \det(T) = 2 \cdot 4 - 3 \cdot 1 = 5.$$

Thus T scales all areas by 5 and preserves orientation.

Example. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ have matrix

$$[T] = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

A direct computation gives $\det(T) = -2$, so T reverses orientation and doubles volume.

If $P : V \to V$ is a change of basis and $S = P^{-1}TP$ - i.e. the representation of T with respect to this new basis, then

$$\det(S) = \det(P^{-1}TP) = \det(T).$$

Thus similar linear operators have the same determinant. In particular, the determinant is a *conjugacy invariant* and depends only on the linear transformation, not on coordinates.

In principle, the determinant $\det(T)$ of a linear transformation captures three central features of a linear transformation:

- **Invertibility:** $\det(T) \neq 0 \iff T$ is bijective.
- **Volume scaling:** $|\det(T)|$ is the n-volume scale factor.
- **Orientation:** $\mathrm{sign}(\det(T))$ records orientation change.

Exercise.

- ▶ Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be diagonal with entries $d_1, \ldots, d_n$. Show that $\det(T) = d_1 \cdots d_n$ and interpret $|d_i|$ as the stretch factor along the i-th axis.
- ▶ Let $R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ be rotation by angle θ. Compute $\det(R_\theta)$ and explain why your answer matches the geometric interpretation.
- ▶ Suppose $H : \mathbb{R}^n \to \mathbb{R}^n$ is the reflection across a hyperplane through the origin. Show that $\det(H) = -1$.

2.2 Characteristic Polynomial

When we choose a basis $\mathcal{B}$ of V and represent T as a matrix $[T]_\mathcal{B}$, the equation $(T - \lambda I)(v) = 0$ becomes

$$([T]_\mathcal{B} - \lambda I)\,[v]_\mathcal{B} = 0.$$

For a nonzero solution, the determinant must vanish:

$$\det([T]_\mathcal{B} - \lambda I) = 0.$$

This determinant is a polynomial in λ of degree $\dim V$, called the **characteristic polynomial**. Its roots are exactly the eigenvalues of T.

2.3 Eigenvalues of Special Transformations

Theorem 2.2. *If $T : V \to V$ is represented in some basis by an upper (or lower) triangular matrix, then its eigenvalues are precisely the diagonal entries of that matrix. Equivalently: the eigenvalues of T are the diagonal entries of $[T]_\mathcal{B}$ for any basis in which T is triangular.*

Example. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by

$$T(x, y, z) = (x + 2y - 2z,\ 3y + 5z,\ -z).$$

Relative to the standard basis, the matrix is $\begin{bmatrix} 1 & 2 & -2 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{bmatrix}$, which is upper triangular. Thus the eigenvalues are $1, 3, -1$.

2.4 Diagonalization as a Coordinate Simplification

Definition 2.3. A linear transformation $T : V \to V$ is said to be **diagonalizable** if there exists a basis $\mathcal{B}$ of V consisting of eigenvectors of T. In this basis, the representation $[T]_\mathcal{B}$ is diagonal, with the eigenvalues of T along the diagonal.

Example. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be given by $T(x, y) = (x + y, 2y)$. Its eigenvalues are $\lambda = 1, 2$ with eigenvectors $(1, 0)$ and $(1, 1)$. Thus, with respect to the basis $\{(1, 0), (1, 1)\}$, the matrix representation is diagonal:

$$[T]_\mathcal{B} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

Theorem 2.3 (Diagonalizability criterion)**.** *A linear transformation $T : V \to V$ is diagonalizable if and only if V has a basis consisting of eigenvectors of T.*

In summary, eigenvalues and eigenvectors are intrinsic to linear transformations. They do not depend on the choice of basis, although their computation is facilitated by choosing one. In an eigenbasis, the transformation is represented by a diagonal matrix, which reveals its action as simple scaling along invariant directions.

2.5 Diagonalization

An important consequence of eigenvalues and eigenvectors is the possibility of simplifying a matrix via diagonalization.

Definition 2.4. A matrix $A \in \mathbb{R}^{n\times n}$ is *diagonalizable* if there exists an invertible matrix P such that

$$P^{-1}AP = D,$$

where D is diagonal. In this case, the diagonal entries of D are the eigenvalues of A, and the columns of P are the corresponding eigenvectors.

Example. Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Take $P = I$, then $P^{-1}AP = A$ itself is diagonal.

For $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$, the eigenvectors are $(1,0)^\top$ and $(1,1)^\top$. Choosing $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, one computes $P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$. Thus A is diagonalizable.

Theorem 2.4 (Diagonalizability criterion)**.** *If an $n \times n$ matrix has n linearly independent eigenvectors, then it is diagonalizable.*

Eigenvalues and eigenvectors thus provide a fundamental way of "decoding" the structure of a linear transformation. They highlight invariant directions, and their algebraic and geometric multiplicities determine whether a matrix can be simplified into a diagonal form.

3 Diagonalization

Definition 3.1. A matrix A is **diagonalizable** if there exists an invertible P such that

$$P^{-1}AP = D$$

with D diagonal.

Theorem 3.1. *If A has n linearly independent eigenvectors, then A is diagonalizable.*

4 Spectral Theorem

Theorem 4.1 (Spectral Theorem)**.** *If A is a real symmetric matrix, then there exists an orthogonal matrix Q such that*

$$Q^\top AQ = \Lambda$$

where Λ is diagonal with the eigenvalues of A on its diagonal.

5 Geometric Interpretation

Eigenvectors are directions preserved by A, scaled by λ.

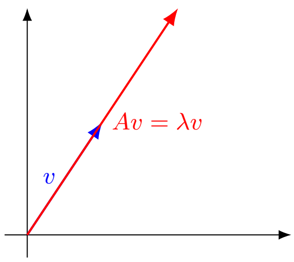

Figure 1: An eigenvector v retains its direction under A, only scaled by λ.

6 Schur Decomposition

Theorem 6.1 (Schur Decomposition)**.** *For any $A \in \mathbb{C}^{n\times n}$, there exists a unitary Q such that*

$$Q^*AQ = T$$

is upper triangular, with eigenvalues of A on the diagonal.